

“IN A WORD THE CORRUPTION AND **WANT OF [TEETH]** IS AS GREAT A **DEFORMITY**, AND OF AS MUCH **PREJUDICE** TO ONE, AS ANYTHING WHATSOEVER CAN BE.”

Charles Allen, British dental practitioner, from *The Operator for the Teeth*, 1685



Painted by Dutch artist Gerrit von Honthorst, *The Tooth Extractor* illustrates the crudities of 17th-century dentistry.



Newton argued that the Moon is subject to the force of Earth's gravity.

IN 1685, EARLY DENTAL PRACTITIONER CHARLES ALLEN

published the first book written in English on dental procedures, *The Operator for the Teeth*. Dentistry had been attempted—with varying degrees of success—since the ancient civilizations, but specific “operators for the teeth” emerged only in the 17th century. These **early dentists gave advice on dental hygiene**, made artificial teeth, and also performed extractions, without anesthetic, using a “pelican”—an instrument so-called because of its resemblance to the bird's bill.

This latter part of the 17th century saw important advances in the classification of life's



Edmond Halley

Although perhaps best known for his work on astronomy, Halley was also a mathematician and geophysicist, and became professor of geometry at Oxford University in England.

diversity. Naturalists catalogued and classified animals and plants based on their structure, often performing painstaking dissections of specimens to do so. Prominent among them was English naturalist **John Ray** (1627–1705), who published the first volume of his treatise *The History of Plants* in 1686, a work relying heavily on his travels in Europe. Ray created a system of classification to organize his catalog and, significantly, **formalized the idea of a species**. He emphasized the importance of reproduction: that seeds sprouting from the same parent plant belong to the same species, even though they may exhibit accidental variations. Ray's concept was to be adopted by generations of naturalists.

Another English naturalist, **Francis Willughby** (1635–72), had studied at Cambridge University, England, under John Ray with whom he collaborated in much of their work on



classification. Following Willughby's premature death in 1672, Ray published his studies posthumously. Willughby's treatise *Ornithology*, which had appeared in 1676, was the first book to take a **scientific approach to the study of birds**. *The History of Fishes*, published in 1686, was another ground-

Historia Plantarum

John Ray's three-volume treatise appeared from 1686 to 1704. He classed plants as either herbs or trees, and distinguished between spore- and seed-bearing plants.

breaking achievement in natural history but sold poorly, which meant that its publisher, the Royal Society, could not afford to fund Isaac Newton's *Principia* a year later.

English mathematician **Edmond Halley**, already known for his astronomical discoveries, also studied the terrestrial atmosphere. In 1686, he suggested that surface winds occurred

because of a pattern of atmospheric circulation that was ultimately driven by heat from the Sun. Tropical warmth at the equator makes air there rise, causing more air to rush in to the region of low pressure. This phenomenon provided the basis for Halley's explanation of the behavior of trade winds and monsoons. At this time he also revisited observations made by other researchers 40 years earlier: that **atmospheric pressure decreases with altitude** (see 1645–54). Halley searched for the quantitative relationship between pressure and altitude, and so established routine use of the **barometer** in practical surveying.

IN THE SUMMER OF 1687, THE ROYAL SOCIETY IN LONDON AUTHORIZED PUBLICATION

of Isaac Newton's *Philosophiae Naturalis Principia Mathematica* (*Mathematical Principles of Natural Philosophy*). In this celebrated book (usually referred to simply as *Principia*), regarded by some as the most important scientific work ever produced,



ISAAC NEWTON
(1642–1727)

Arguably the greatest of all mathematicians, Newton founded classical mechanics, invented calculus, and made breakthrough discoveries about gravity and light. He studied at Cambridge University, England, where he became a professor of mathematics. After reforming the coinage of the Royal Mint, he was elected president of the Royal Society in 1703 and knighted in 1705.

“IN ORDER THAT AN **INVENTORY OF PLANTS** MAY BE BEGUN... WE MUST... DISCOVER **CRITERIA...** FOR **DISTINGUISHING** WHAT ARE CALLED '**SPECIES.**' ”

John Ray, from *Historia Plantarum*, 1686

1685 Englishman Charles Allen publishes *The Operator for the Teeth*, an early work on dentistry

1686 John Ray publishes the first volume of *History of Plants*, containing the first definition of the term “species”

1686 Edmond Halley explains atmospheric circulation and the link between atmospheric pressure and altitude

July 5, 1687 Isaac Newton publishes his *Principia*, the foundation for classical mechanics

1686 Francis Willughby's *History of Fishes* is published posthumously